

P/G Theory — innovation frontiers (merged set)

P/G 理论 —— 创新疆域（合并版）

Nine directions in which the P/G build-method could open new ground in economics. Directions 1–6 keep the sharper framing of the refined draft; 7–9 add the deontic, ecological, and diversity directions. Each names the idea and its hardest obstacle. A recommendation on where to bet closes the document.

P/G 的构建方法可能在经济学中开辟新疆域的九个方向。第 1–6 项保留精炼草稿中更锋利的表述;第 7–9 项补上义务论、生态与多样性三个方向。每一项都给出核心想法及其最难的障碍。文末以一段“该押注哪个”的建议收尾。

1 · Relational macroeconomics: give G its own equations of motion / 关系宏观经济学:赋予 G 自身的运动方程

Economics usually treats trust and social capital as a slow, exogenous parameter. The innovation is to make G an endogenous, fast-moving state variable with its own dynamics — accumulation, destruction, sign-flips — that feeds back into P. This could give a micro-founded formalization of Keynes's “animal spirits” and the Akerlof–Shiller confidence cycle: booms and busts as phase changes in the relationship field, not just price shocks. **Hardest obstacle:** writing G's laws of motion so they are disciplined by data rather than freely invented.

经济学通常把信任与社会资本当作一个缓慢的、外生的参数。创新在于:把 G

变成一个内生、快速变动、拥有自身动态(积累、破坏、正负翻转)的状态变量,并反馈回

P。这可能为凯恩斯的“动物精神”和 Akerlof–Shiller

的信心周期提供微观基础的形式化:把繁荣与萧条理解为关系场中的相变,而不只是价格冲击。**最难的障碍:**写出 G 的运动方程,使其受数据约束,而非凭空臆造。

2 · Relational welfare economics: put G inside the welfare function / 关系福利经济学:把 G 放进福利函数

Current social welfare functions are functions of individual utilities — essentially P. A normatively new object is a welfare criterion that also includes the state of G: a tax, platform, or market design scored not only on the distribution of P but on whether it builds or erodes trust, cohesion, and cooperation potential. Mechanism design gains a new target — relational welfare. **Hardest obstacle:** the theory itself warns that descriptive power is not normative authority, so you must answer “who defines a healthy G,” paired with non-overridable hard constraints, or it becomes dangerous social engineering.

当前的社会福利函数是个体效用的函数——本质上就是 P。一个在规范意义上全新的对象,是一个同时包含 G 状态的福利标准:对一项税收、平台或市场设计的评分,不仅看 P 的分布,还看它是建立还是侵蚀了信任、凝聚力与合作潜能。机制设计因此获得一个新目标——关系福利。**最难的障碍:**该理论自身警告“描述能力不等于规范权威”,因此你必须回答“由谁来界定健康的 G”,并配以不可推翻的硬约束,否则它会沦为危险的社会工程。

3 · Tipping-point economics with relational microfoundations / 具关系微观基础的临界点经济学

The model's threshold-gated mode switch — like bacterial quorum sensing, where G crosses a threshold and behaviour flips — is a phase-transition mechanism. The new lens: bank runs, cooperation collapse, market freezes, and polarization as bifurcations in the G-field, triggered when P crosses personality or context thresholds. This connects Schelling's tipping points and catastrophe theory to a relational microfoundation. **Hardest obstacle:** identifying the flip point empirically, rather than only recognizing it in hindsight.

模型中阈值触发的模式切换——类似细菌的群体感应:G

越过阈值,行为随之翻转——是一种相变机制。新的视角:把银行挤兑、合作崩塌、市场冻结和极化,理解为 G 场中的分岔,由 P 越过性格或情境阈值所触发。这把 Schelling

的临界点与突变论连接到一个关系微观基础上。**最难的障碍**:在经验上识别出翻转点,而不是只能事后才看清它。

4 · A measurement science for relational preferences (the identification program) / 关系偏好的测量科学(识别纲领)

Social preferences are currently inferred only indirectly. A genuinely new experimental and empirical branch would design experiments and markets that identify G-weights separately from P-weights — revealing how much people will pay to change a relationship versus a resource. This is also the theory's number-one bottleneck: solve G-identification and the first three fields become tractable. **Hardest obstacle**: identifiability — too many free parameters invite overfitting, so it demands clever design and exclusion restrictions.

社会偏好目前只能被间接推断。一个真正全新的实验与经验分支,将设计能把 G 权重与 P 权重分别识别出来的实验和市场——揭示人们为改变一段关系、相对于改变一项资源,各自愿意付出多少。这也是该理论的头号瓶颈:解决 G 的识别问题,前面三个领域便都变得可行。**最难的障碍**:可识别性——自由参数过多会招致过拟合,因此需要精巧的设计与排除性约束。

5 · The economics of AI agents (most natural fit, possibly highest impact) / AI 主体的经济学(最契合、可能影响最大)

As AI agents begin transacting — agentic commerce — they need exactly what P/G supplies: relational state (trust, reputation, obligation), bounded look-ahead, and hard constraints. Designing markets, contracts, reputation systems, and safety boundaries for populations of P/G-style AI agents is a near-future field, and it directly echoes the theory's "life-compatible / cooperative AI" stance. **Hardest obstacle**: this is also a safety problem — an interpretable relational kernel is an asset, but you must stop agents from treating G as a manipulable attack surface.

当 AI 主体开始交易——主体化商务——它们所需的恰恰是 P/G 所提供的:关系状态(信任、声誉、义务)、有限前瞻,以及硬约束。为 P/G 式 AI 主体群体设计市场、合约、声誉系统和安全边界,是一个近在眼前的领域,并直接呼应该理论"兼容生命/合作型 AI"的立场。**最难的障碍**:这同时也是一个安全问题——可解释的关系内核是一项资产,但你必须防止主体把 G 当作可被操纵的攻击面。

6 · Scale-free recursive P/G (boldest, most speculative) / 尺度无关的递归 P/G(最大胆、最具推测性)

The most ambitious claim is that the same core runs at every level of complexity. Ported to economics: one P/G loop describing individual → firm → industry → nation, with thresholds switching on new capacities — planning, norms, institutions — at each level. The prize is a scale-free behavioural economics that runs one recursive relational operator across all levels of aggregation, sidestepping the usual micro-to-macro aggregation problem. **Hardest obstacle**: this slides very easily into an unfalsifiable metaphor; you would need a concrete model at two adjacent scales that can make a wrong prediction.

最雄心勃勃的主张是:同一个核心在每一个复杂度层级上运行。移植到经济学:一个 P/G 循环描述个体 → 企业 → 行业 → 国家,在每一层由阈值开启新的能力——规划、规范、制度。其奖赏是一种尺度无关的行为经济学,用一个递归的关系算子贯穿所有加总层级,绕开通常的微观到宏观加总难题。**最难的障碍**:这极易滑向一个不可证伪的隐喻;你需要在两个相邻尺度上建立一个能做出"可被证错"预测的具体模型。

7 · Deontic economics: non-tradeable protected values / 义务论经济学:不可交易的受保护价值

Standard economics folds everything into one objective, so rights become shadow prices and in effect everything has a price. P/G keeps hard constraints categorically outside the value function — zero probability on actions outside the safe set — giving a formal way to represent protected or sacred values and taboo trade-offs. The new field: economic models in which consent, rights, and irreversibility are inviolable constraints rather than priced

costs, directly relevant to regulation and to AI agents that must not be bribed past their limits. **Hardest obstacle:** deciding which values are hard versus soft, and resolving unavoidable conflicts between hard constraints.

标准经济学把一切都收进单一目标,于是权利变成影子价格,实际上"万物皆有价"。P/G 把硬约束绝对地置于价值函数之外——对安全集合之外的行为赋予零概率——从而提供了一种表示受保护或神圣价值,以及禁忌权衡的正式方式。新领域:把同意、权利和不可逆性作为不可侵犯的约束(而非可定价成本)的经济模型,这与监管直接相关,也与"不能被收买、越过底线"的 AI

主体直接相关。 **最难的障碍:**判定哪些价值是硬约束、哪些是软约束,并解决硬约束之间不可避免的冲突。

8 · Life-compatible, multi-scale economics (ecological & intergenerational G) / 兼容生命的多尺度经济学(生态与跨代的 G)

Externalities and future generations enter standard models awkwardly, through discounting wars and Pigouvian patches. P/G lets nested G — reaching up to ecosystem and future life — be native weights inside an agent's valuation, so ecological and intergenerational stakes are internal to the agent rather than an external correction. The new field: studying whether economies populated by agents with non-zero ecosystem- and future-G weights avoid the commons tragedies that pure self-payoff agents generate, and designing institutions that cultivate such weights. **Hardest obstacle:** where higher-scale weights come from, and whether they can be sustained against free-riding.

外部性与后代在标准模型中进入得很别扭——靠折扣率之争和庇古式补丁。P/G 让嵌套的 G(一直延伸到生态系统与未来生命)成为主体估值中的原生权重,于是生态与跨代利害是主体内部的一部分,而非外部修正。新领域:研究由"对生态与未来 G 赋予非零权重"的主体所构成的经济体,能否避免纯自利主体所制造的公地悲剧,并设计培育此类权重的制度。 **最难的障碍:**更高尺度的权重从何而来,以及它们能否在搭便车压力下得以维持。

9 · Diversity economics: the population of types as the primitive / 多样性经济学:以类型分布为基本对象

Representative-agent and even heterogeneous-agent macro usually vary only one or two parameters; the composition of behavioural types is rarely the central object. The Behavioural Diversity Principle makes the distribution of weight-vectors and time-horizons the fundamental state, so one can ask how the mix of types determines stability. The new field: an ecology of economic types, importing evolutionary-stability and biodiversity-style reasoning into market composition — which mixes are resilient, which collapse, and how the mix itself drifts. **Hardest obstacle:** identifying the type distribution from data, and modelling the dynamics of type composition.

代表性主体、乃至异质性主体的宏观模型,通常只改变一两个参数;行为类型的"构成"很少成为核心对象。行为多样性原则把权重向量与时间视野的分布当作基本状态,于是可以追问:类型的配比如何决定稳定性。新领域:一种"经济类型的生态学",把演化稳定性与生物多样性式的推理引入市场构成——哪些配比有韧性、哪些会崩溃,以及这种配比本身如何漂移。 **最难的障碍:**从数据中识别类型分布,并对类型构成的动态进行建模。

Which to bet on? / 该押注哪一个?

If the goal is a new field that actually stands up, bet on #4 (relational measurement) plus #1 (relational macro): #4 is the key, #1 is the door it opens, and together they are falsifiable and publishable. #5 (AI-agent economics) has the highest impact and the best timing, but it is more engineering-and-safety than pure theory — #7 (deontic) is its natural safety companion. #2 and #3 are elegant but carry normative and identification risks respectively. #6 (scale-free recursion) and #8 (ecological / intergenerational) are high-risk, high-reward and stay philosophical until they make a concrete, falsifiable prediction; #9 (diversity) pairs naturally with the #4 measurement program. In one line: this theory's real new continent is turning $\Delta P / \Delta G$ into a measurable scientific object — every field above only appears once you take that step.

如果目标是一个真正立得住的新领域,押注第 4 项(关系测量)加第 1 项(关系宏观):第 4 项是钥匙,第 1 项是它打开的门,两者合在一起可证伪、可发表。第 5 项(AI

主体经济学)影响最大、时机最佳,但它更偏工程与安全,而非纯理论——第 7 项(义务论)是它天然的安全搭档。第 2、第 3 项优雅,但分别带有规范与识别上的风险。第 6 项(尺度无关递归)与第 8 项(生态/跨代)高风险、高回报,在做出具体的、可证伪的预测之前仍停留在哲学层面;第 9 项(多样性)与第 4 项的测量纲领天然配对。一句话:这个理论真正的新大陆,是把 $\Delta P / \Delta G$ 变成一个可测量的科学对象——上面每一个领域,都只有在你迈出这一步之后才会出现。